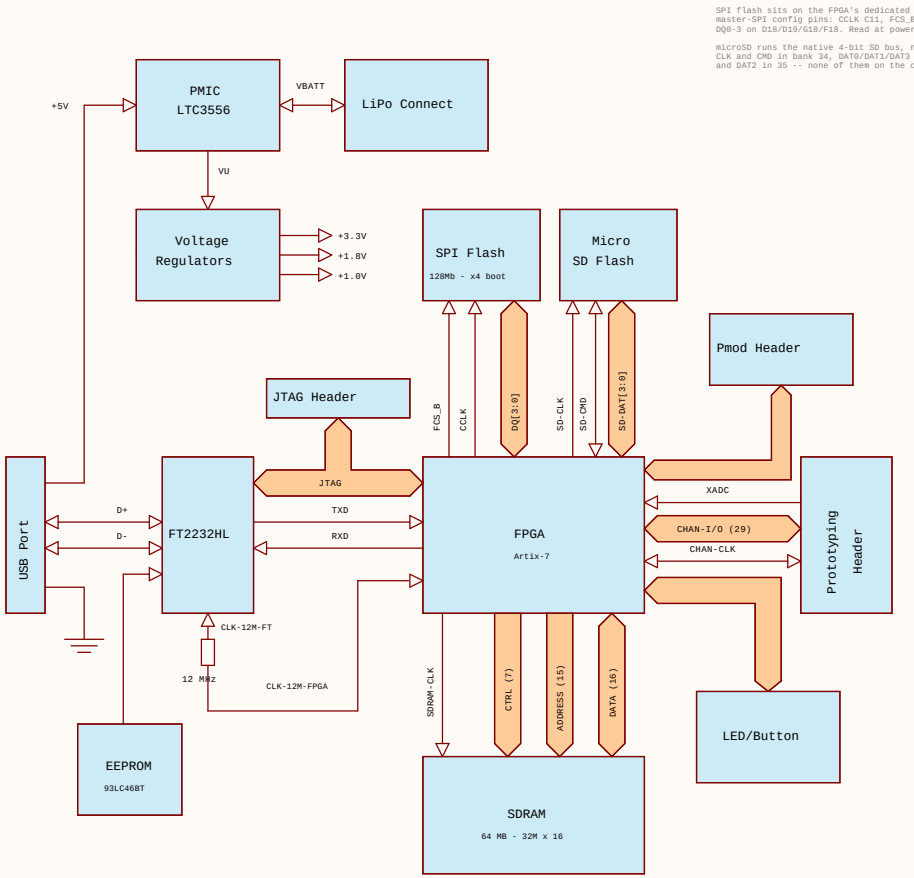
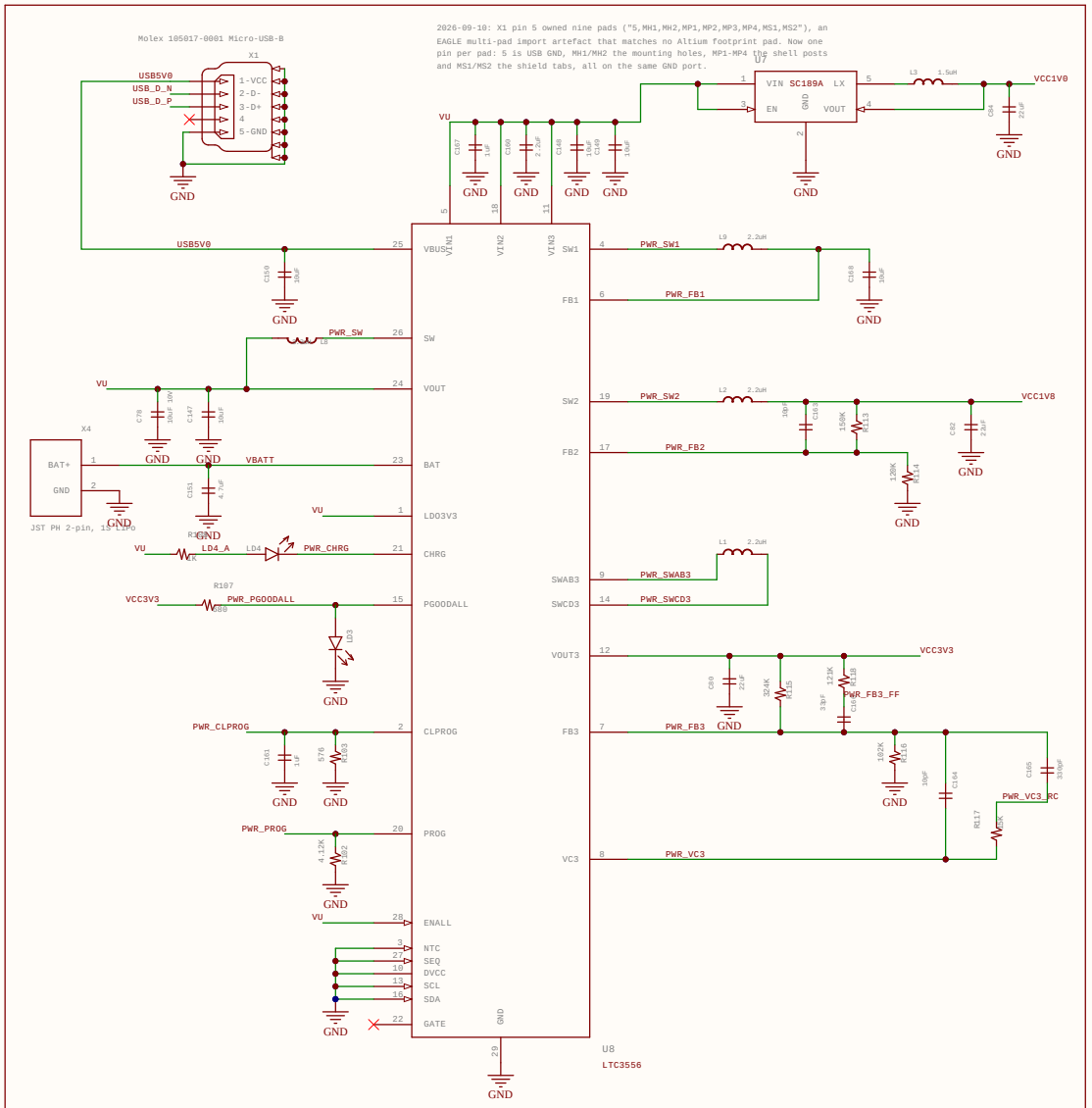


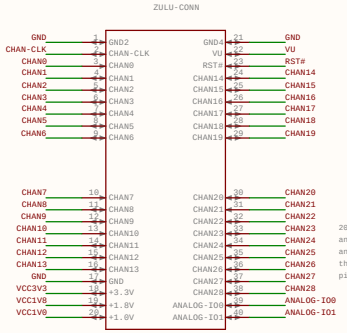
Block Diagram			
PROJECT:	**zulu_a7**	REV:	9/2/2026 1:29:21 PM
AUTHOR:	Christian Tambe		Sheet: **1**



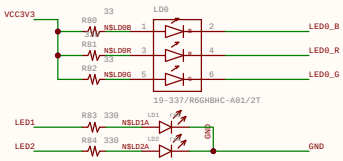
2026-11-10: U8 bq24232 + U5 TPS63802 + U6 SC189L -> one Analog Devices LTC3556 (USB PowerPath + charger 248 mA, Buck2 VCC1V8 400 mA, buck-boost VCC3V3 1 A); VCC1V0 stays on U7 SC189A (1.5 A: XPE puts a half-full XC7A35T at ~0.4 A). U7 starts on ENALL; Buck1 runs unloaded as a sequencing slot (SEQ = GND), so 1V0 -> 1V8 -> 3V3. I2C off (DVCC/SCL/SDA = GND): USB input limit = 1x mode set by R103 = 576R: ~465 mA typ, 489 mA max (500 mA USB). NTC grounded, GATE open. LD3 = rails good, LD4 = charging.

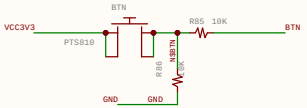
Power Supplies			
PROJECT:	zulu_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR:	Christian Tambe		Sheet: 2

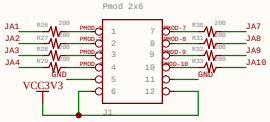
X2: 40 pins on a 2.54 mm grid, rows 22.86 mm (0.900 in) apart, numbered 1-40 straight through from the right and skipping both landings. The two rows are mirror images.
TOP ROW, x 59.69 down to 1.27: GND (pin 1), CHAN-CLK (2), CHAN0..CHAN6 (3-9), then CHAN7..CHAN13 (10-16), GND, +3.3V, +1.8V, +1.0V (17-20).
BOTTOM ROW, x 59.69 down to 1.27: GND, VU, RST# (pins 21-23), CHAN14..CHAN16 (24-26), CHAN17..CHAN19 (27-29), then CHAN20..CHAN28 (30-38), ANALOG-IO0, ANALOG-IO1 (39, 40).
FOUR positions in each row -- x 36.83, 34.29, 31.75, 29.21, between pins 9 and 10 above and between pins 29 and 30 below -- are RESERVED, not spare: the edge-mounted micro-USB X1 lands on the top row (7.00 mm of copper), the top-side LiPo header X4 (JST B2B-PH-SM4-TB, 7.95 mm body) lands on the bottom row directly opposite it.



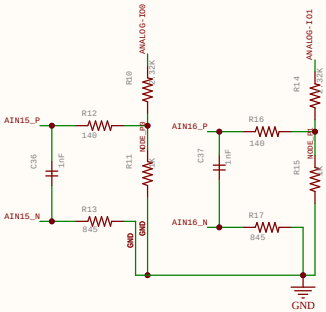
2026-09-10: BTN pads 1-2 are one switch terminal and 3-4 the other. They were one pin each ("1,2" and "3,4"), an EAGLE multi-pad import artefact that matches no Altium footprint pad; now four pins, each pad on its partner's net.







Digilent 12-pin Pmod (JA). Pins 1-4 and 7-10 are signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the Pmod spec puts bypassing on the module, and VCC3V3 already carries ~170uF. The Cmod A7 has none at its Pmod either.
JA1-JA4 (the top row, the half a Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16.
JA7 is C15, also bank 16. JA8/JA9/JA10 are W19/N2/W7 -- banks 14/35/34 -- because nothing else was free. All five banks are VCCO=3.3V, so this costs timing coherence, not levels.



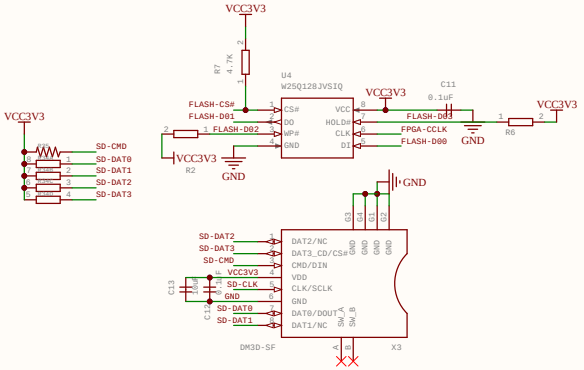
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General IO			
PROJECT: zulu_a7	REV:	9/2/2026 1:29:21 PM	
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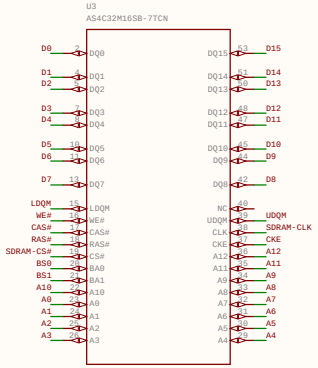
MASTER SPI BOOT (x4). M2:M1:M0 = 0:0:1 -- R20 pulls M0 to VCC3V3, R21/R22 pull M1/M2 to GND -- so at power-up the FPGA drives CCLK itself and clocks its bitstream out of U4 with no USB and no host present. That mode can only use the dedicated configuration pins: CCLK C11, FCS_B K19, D00_MOSI D18, D01_DIN D19, D02 G18, D03 F18. No other ball will do, which is why SDRAM A4 had to move off K19 (it is now on D17, still bank 14). After DONE the six pins revert to ordinary user I/O. R2A/R2B keep WP#/HOLD# high through the x1 command phase and through the float window before the FPGA drives D02/D03 (PUDC_B is tied high, so the FPGA has no internal pull-ups). R2D keeps FCS_B deselected.
Vivado: set_property CONFIG_MODE SPIx4 [current_design] set_property BITSTREAM.CONFIG.SPI_BUSWIDTH 4 [current_design]
The microSD is deliberately NOT on this bus any more: an inserted card sits in native SD mode watching bitstream traffic go past on what it believes is CMD, and one 48-bit window with a valid CRC7 is enough to make it answer and drive the line the FPGA is reading. X3 now has its own CLK/CMD/DAT0 on R2/T1/T2, the bank-34 balls the flash just vacated, with R9 damping DAT0 where the socket stub is.

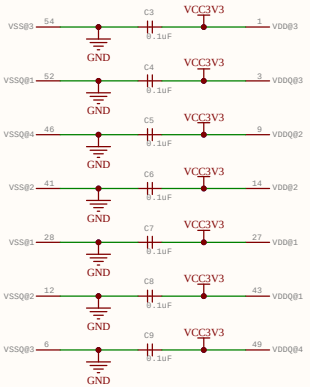
U3 = AS4C32M16SB-7TCN: 512Mbit (32M x 16) = 64MB. Pin-for-pin and dimensionally identical to the 256Mb AS4C16M16SA-6TIN it replaces -- all 54 pins and every TSOP-II dimension checked against both datasheets, and pin 40 is NC on both.
ONE DIFFERENCE, AND IT IS NOT A PIN: the column address is A0-A9 (1024 columns), not A0-A8 (512). The memory controller must be set to 10 column bits, or it reaches only half the array. Row A0-A12, banks BA0/BA1 and refresh (8192 cycles / 64ms) are unchanged.
-7TCN since 2026-09-08 (-6TIN dry): 143 MHz max at CL3, tRCD/tRP 21, tRC 63 ns, TA 0-70 C, IDD1 max 110 mA.



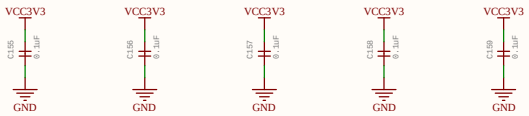
microSD runs the NATIVE 4-BIT SD BUS, not SPI. CLK, CMD and DAT0-DAT3 -- six signals, all in bank 34 byte group T0 (R2 T1 T2 U1 V2 W2), so they can be constrained as one bus. DAT1/DAT2 came from prototyping-header CHAN1/CHAN0; DAT3 is the old SPI chip-select, which needs no new pin. R34/R35 are the 4.7k pull-ups the SD spec requires on CMD and DAT0-DAT3. R9 (100R on DAT0 only) was removed: one damped line out of four is a skew, not termination.

2026-09-06: X3 is now the Hirose DM3D-SF push-pull socket (HRS 609-0025-8), DM3 catalog p9. Same 8-contact pinout; shell pads G1-G4 to GND; card-detect switch A/B (closed with card) left open. Needs its own footprint on the PCB.

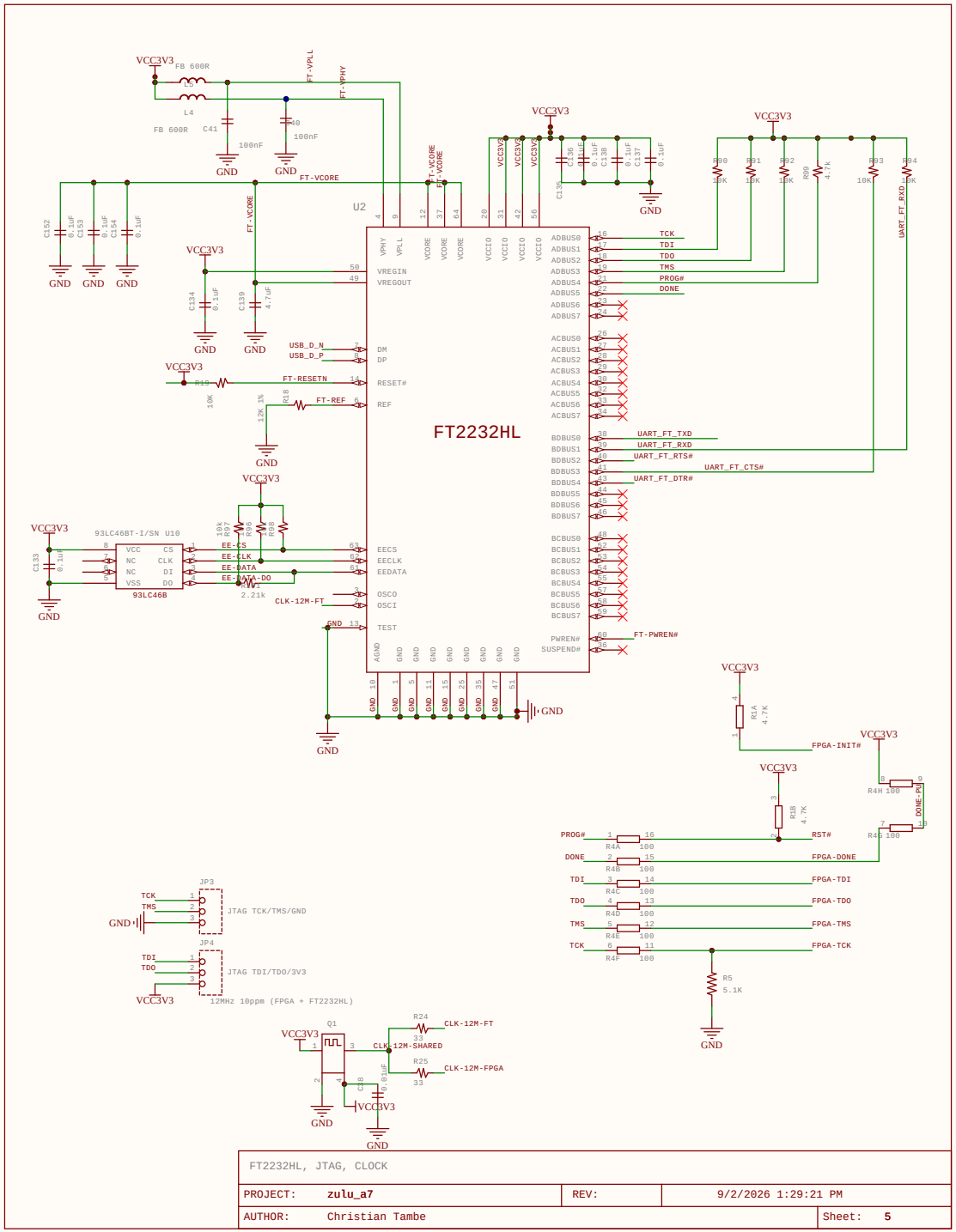


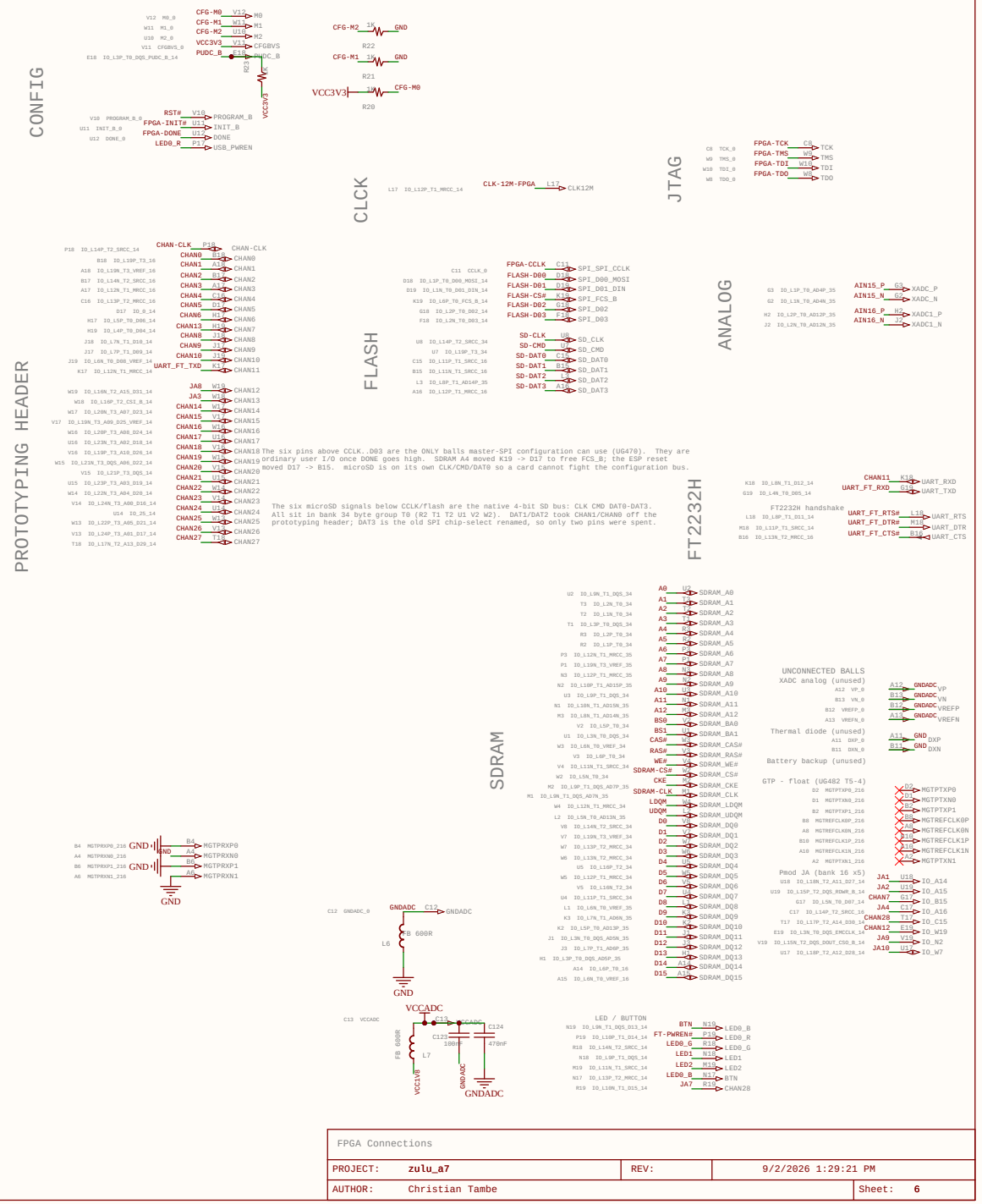


U3 DECOUPLING -- each cap bridges one supply pin to the ground pin OPPOSITE it. In the 54-pin TSOP-II, pin n faces pin 55-n, and every supply pin faces a ground of its own domain: VDD 1/14/27 to VSS 54/41/28, VDDQ 3/9/43/49 to VSSQ 52/46/12/6. Alliance isolates VDDQ from VDD for DQ noise immunity -- LAYOUT MUST KEEP THE PAIRS: each cap on the underside spanning its two pins, own vias, VDDQ returning to VSSQ and VDD to VSS. C10 is bulk and pairs with nothing. Do NOT split the ground plane.



Memory			
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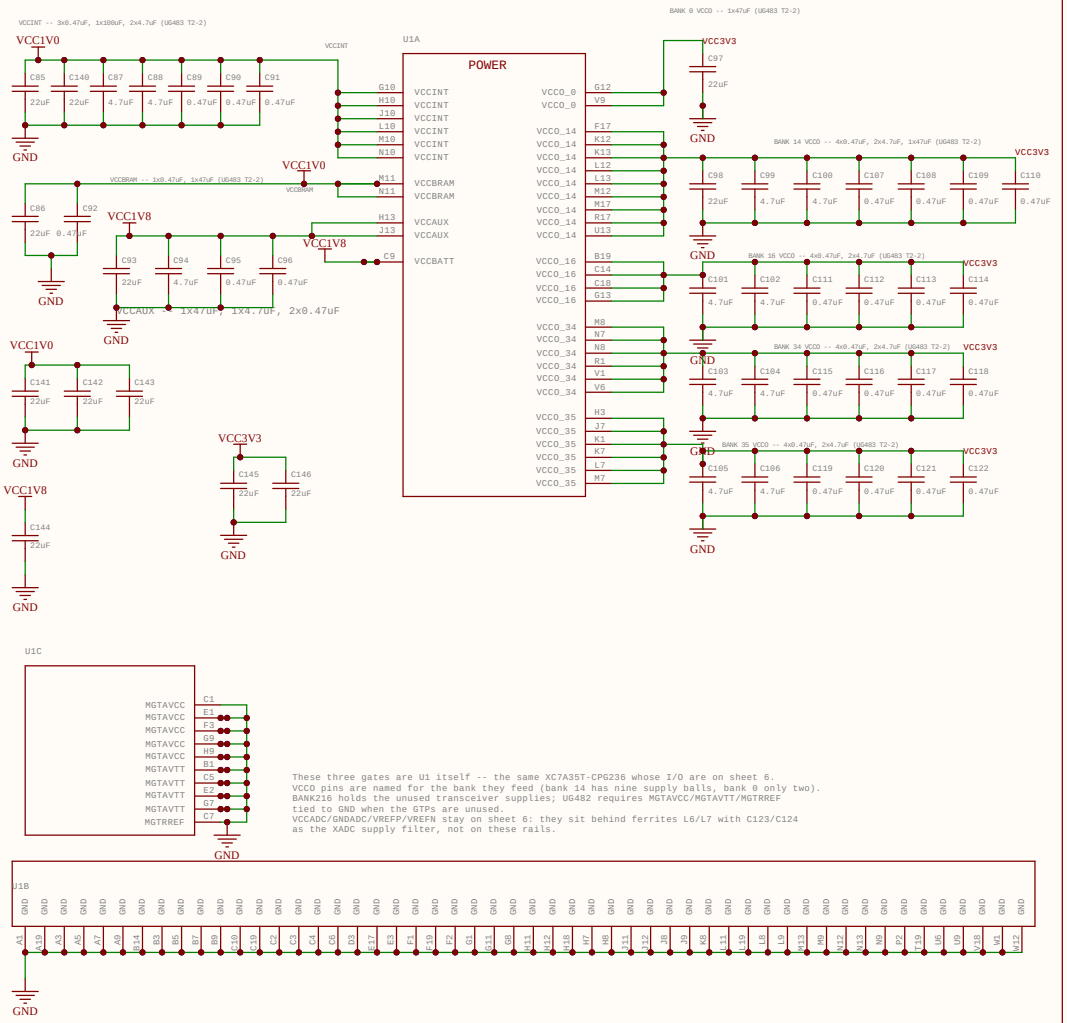

FT2232HL
U2
93LC46BT-I/SN U10
93LC46B
JP3
JTAG TCK/TMS/GND
JP4
JTAG TDI/TDO/3V3
12MHz 10ppm (FPGA + FT2232HL)
Q1
FT2232HL, JTAG, CLOCK
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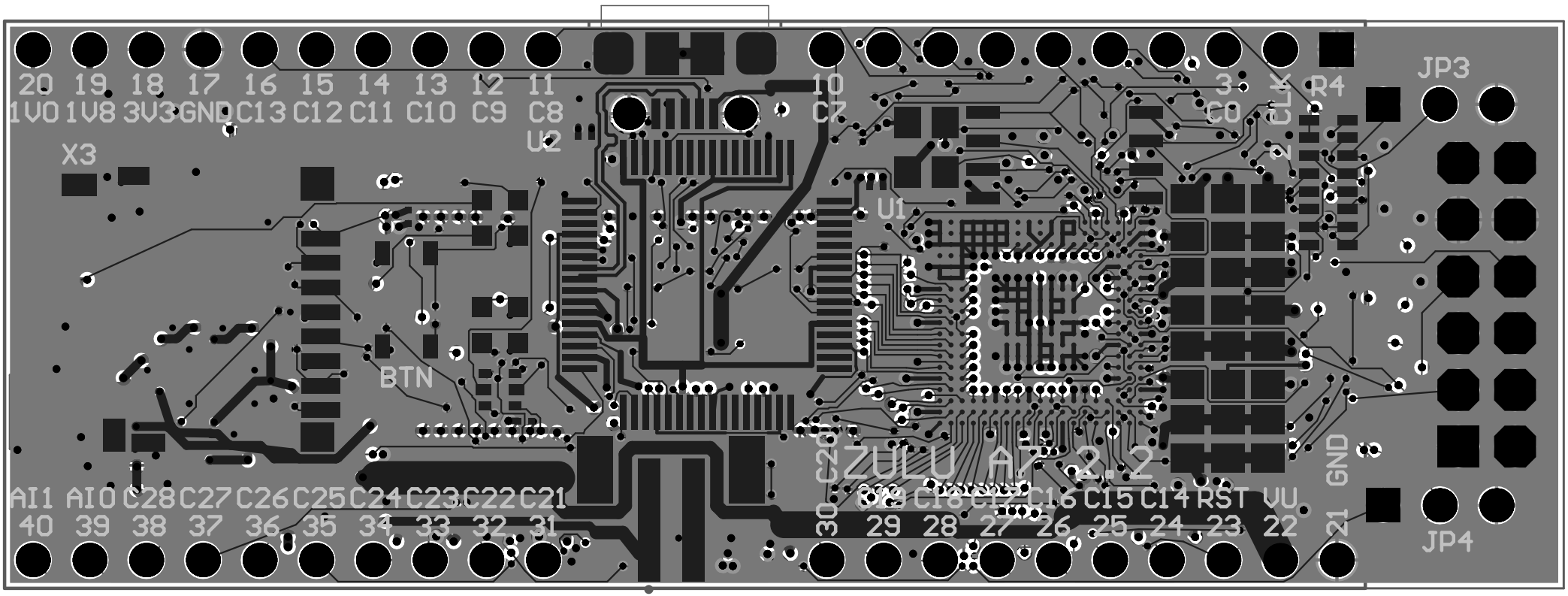
The six pins above CCLK..D03 are the ONLY balls master-SPI configuration can use (UG470). They are ordinary user I/O once DONE goes high. SDRAM A4 moved K19 -> D17 to free FCS_B; the ESP reset moved D17 -> B15. microSD is on its own CLK/CMD/DAT0 so a card cannot fight the configuration bus.

The six microSD signals below CCLK/flash are the native 4-bit SD bus: CLK CMD DAT0-DAT3. All sit in bank 34 byte group T0 (R2 T1 T2 U1 V2 W2). DAT1/DAT2 took CHAN1/CHAN0 off the prototyping header; DAT3 is the old SPI chip-select renamed, so only two pins were spent.

FPGA Connections			
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FPGA POWER			
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20 19 18 17 16 15 14 13 12 11
1V0 1V8 3V3 GND C13 C12 C11 C10 C9 C8
X3
U2
10
C7
U1
3
C0
CLK
R4
JP3
BTN
ZULU A7 2.2
C26
AI1 AI0 C28 C27 C26 C25 C24 C23 C22 C21
40 39 38 37 36 35 34 33 32 31
30
29 28 27 26 25 24 23 22
21
GND
JP4

CC BY SA